

Advanced Machine Learning



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Administrative

- seminar 3 class today, 4 exercises
- seminar 4 starts after the Easter holiday
- today you get the assignment

$\text{VCdim}(\mathcal{H}S_0^n)$

Consider $\mathcal{H} = \mathcal{H}S^n$ be the set of halfspaces (linear classifiers) in $\mathbf{R}^n$

$$\mathcal{H} = \mathcal{H}S^n = \{h_{w,b}: \mathbf{R}^n \rightarrow \{-1, 1\}, h_{w,b}(x) = \text{sign}\left(\sum_{i=1}^n w_i x_i + b\right) \mid w \in \mathbf{R}^n, b \in \mathbf{R}\}$$

Consider label -1 to correspond to label 0, they are basically the same.

For $n = 2$ we have:

$$\mathcal{H}S^2 = \mathcal{H}_{\text{lines}} = \{h_{a,b,c}: \mathbf{R}^2 \rightarrow \{0, 1\}, h_{a,b,c}((x,y)) = \mathbf{1}_{[ax+by+c>0]}((x,y)), a, b, c \in \mathbf{R}\}$$

Let us restrict our attention to “homogenous” linear classifiers, the ones that go through origin, $b = 0$.

$$\mathcal{H}S_0^n = \{h_{w,0}: \mathbf{R}^n \rightarrow \{-1, 1\}, h_{w,0}(x) = \text{sign}\left(\sum_{i=1}^n w_i x_i\right) \mid w \in \mathbf{R}^n\}$$

What is the $\text{VCdim}(\mathcal{H}S_0^n)$?

VCdim($\mathcal{HS}_0^n$)

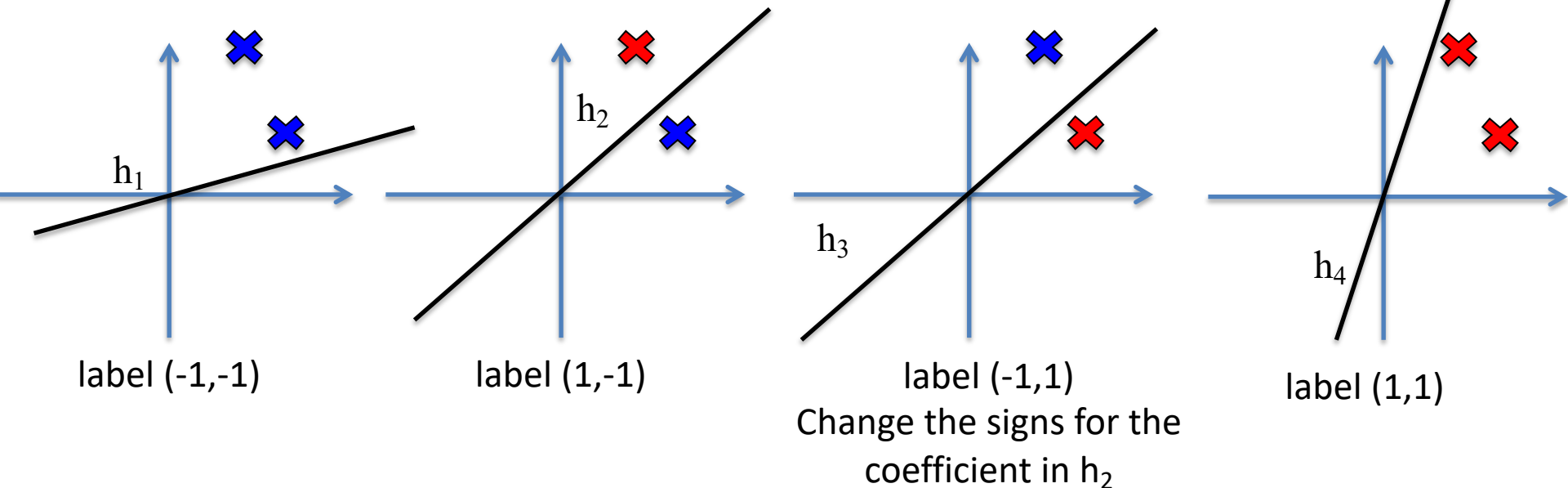
$$\mathcal{HS}_0^n = \{h_{w,0}: \mathbf{R}^n \rightarrow \{-1, 1\}, h_{w,0}(x) = \text{sign}\left(\sum_{i=1}^n w_i x_i\right) \mid w \in \mathbf{R}^n\}$$

For $n = 2$ we have:

$$\mathcal{HS}_0^2 = \{h_{w_1,w_2}: \mathbf{R}^2 \rightarrow \{-1, 1\}, h_{w_1,w_2}(x) = \text{sign}(w_1 x_1 + w_2 x_2) \mid (w_1, w_2) \in \mathbf{R}^2\}$$

What is the VCdim($\mathcal{HS}_0^2$) ?

$\mathcal{HS}_0^2$ shatters any set A of two different points (non-colinear with the origin).



Does $\mathcal{HS}_0^2$ shatter a set A of three points?

Difficult to reason geometrically... choose the algebraic proof.

VCdim($\mathcal{HS}_0^n$)

We will show that $\text{VCdim}(\mathcal{HS}_0^n) = n$.

Proof: 1st part

We first show that $\text{VCdim}(\mathcal{HS}_0^n) \geq n$.

We find a set A consisting of n points in $\mathbf{R}^n$ that is shattered by $\mathcal{HS}_0^n$.

Take $A = \{e_1, e_2, \dots, e_n\}$ to be the orthonormal basis of $\mathbf{R}^n$.

$e_1 = (1, 0, 0, \dots, 0)$; $e_2 = (0, 1, 0, \dots, 0)$; $\dots$; $e_n = (0, 0, 0, \dots, 1)$

We want to prove that $\mathcal{HS}_0^n$ shatters A , so that $\text{VCdim}(\mathcal{HS}_0^n) \geq n$. This is equivalent to prove that for every $B \subseteq A$, there is a function $h_B \in \mathcal{HS}_0^n$ such that h_B gives label +1 to all elements in B and label -1 to all elements of $A \setminus B$.

Pick B subset of A , $B \subseteq \{e_1, e_2, \dots, e_n\}$. Choose $w = (w_1, w_2, \dots, w_n)$ such that:

$$w_i = \begin{cases} 1, & \text{if } e_i \in B \\ -1, & \text{if } e_i \notin B \end{cases}$$

Then, $h_B(e_i) = \text{sign}(\langle w, e_i \rangle) = w_i$ will generate the labels +1 for elements in B , -1 for elements not in B

VCdim($\mathcal{H}S_0^n$)

Proof: 2nd part

We now show that $\text{VCdim}(\mathcal{H}S_0^n) < n + 1$.

We will prove that given any set $A = \{x_1, x_2, \dots, x_{n+1}\}$ of $n + 1$ points in $\mathbf{R}^n$, A cannot be shattered by $\mathcal{H}S_0^n$.

The points $\{x_1, x_2, \dots, x_{n+1}\}$ “live” in $\mathbf{R}^n$, a vector space with dimension n . So, $\{x_1, x_2, \dots, x_{n+1}\}$ are linearly dependent and there exist coefficients $a_1, a_2, \dots, a_{n+1}$ not all of them 0 such that:

$$\sum_{i=1}^{n+1} a_i x_i = 0$$

Take $P \subseteq \{1, 2, \dots, n+1\}$ the set of strictly positive coefficients a_i and $N \subseteq \{1, 2, \dots, n+1\}$ the set of negative coefficients of a_i . So we have:

$$\sum_{i \in P} a_i x_i = \sum_{j \in N} |a_j| x_j$$

VCdim($\mathcal{HS}_0^n$)

Take $P \subseteq \{1, 2, \dots, n+1\}$ the set of strictly positive coefficients a_i and $N \subseteq \{1, 2, \dots, n+1\}$ the set of negative coefficients of a_i . Both P and N cannot be at the same time empty. So we have:

$$\sum_{i \in P} a_i x_i = \sum_{j \in N} |a_j| x_j$$

Assume that A is shattered by $\mathcal{HS}_0^n$ and take $B = \{x_i \mid i \in P\}$. In particular, there exist h_B such that it realizes the label consisting of +1 for all $x_i \in B$ and -1 for all $x_i \notin B$.

So, we have that $h_B(x_i) = 1$, if $x_i \in B$, meaning that $\langle w_B, x_i \rangle \geq 0$ if $x_i \in B$ and $h_B(x_i) = -1$, if $x_i \notin B$, meaning that $\langle w_B, x_i \rangle < 0$ if $x_i \notin B$

So, we have that
$$h_B\left(\sum_{i \in P} a_i x_i\right) = \text{sign}\left(\left\langle w_B, \sum_{i \in P} a_i x_i \right\rangle\right) = \text{sign}\left(\sum_{i \in P} a_i \langle w_B, x_i \rangle\right)$$

But $a_i > 0$ (because $i \in P$) and also $\langle w_B, x_i \rangle \geq 0$ as $x_i \in B$, so we obtain that:

$$h_B\left(\sum_{i \in P} a_i x_i\right) = \text{sign}\left(\left\langle w_B, \sum_{i \in P} a_i x_i \right\rangle\right) = \text{sign}\left(\sum_{i \in P} a_i \langle w_B, x_i \rangle\right) = 1$$

VCdim($\mathcal{HS}_0^n$)

Take $P \subseteq \{1, 2, \dots, n+1\}$ the set of strictly positive coefficients a_i and $N \subseteq \{1, 2, \dots, n+1\}$ the set of negative coefficients of a_i . Both P and N cannot be at the same time empty. So we have:

$$\sum_{i \in P} a_i x_i = \sum_{j \in N} |a_j| x_j$$

Assume that A is shattered by $\mathcal{HS}_0^n$ and take $B = \{x_i \mid i \in P\}$. In particular, there exist h_B such that it realizes the label consisting of +1 for all $x_i \in B$ and -1 for all $x_i \notin B$.

So, we have that $h_B(x_i) = 1$, if $x_i \in B$, meaning that $\langle w_B, x_i \rangle \geq 0$ if $x_i \in B$ and $h_B(x_i) = -1$, if $x_i \notin B$, meaning that $\langle w_B, x_i \rangle < 0$ if $x_i \notin B$

On the other hand, we have that
$$h_B \left(\sum_{j \in N} |a_j| x_j \right) = \text{sign} \left(\left\langle w_B, \sum_{j \in N} |a_j| x_j \right\rangle \right) = \text{sign} \left(\sum_{j \in N} |a_j| \langle w_B, x_j \rangle \right)$$

But $|a_j| > 0$ and also $\langle w_B, x_j \rangle < 0$ as $x_j \notin B$, so we obtain that:

$$h_B \left(\sum_{j \in N} |a_j| x_j \right) = \text{sign} \left(\left\langle w_B, \sum_{j \in N} |a_j| x_j \right\rangle \right) = \text{sign} \left(\sum_{j \in N} |a_j| \langle w_B, x_j \rangle \right) = -1$$

VCdim($\mathcal{HS}_0^n$)

Take $P \subseteq \{1, 2, \dots, n+1\}$ the set of strictly positive coefficients a_i and $N \subseteq \{1, 2, \dots, n+1\}$ the set of negative coefficients of a_i . Both P and N cannot be at the same time empty. So we have:

$$\sum_{i \in P} a_i x_i = \sum_{j \in N} |a_j| x_j$$

Assume that A is shattered by $\mathcal{HS}_0^n$ and take $B = \{x_i \mid i \in P\}$. In particular, there exist h_B such that it realizes the label consisting of +1 for all $x_i \in B$ and -1 for all $x_i \notin B$. So $h_B(x_i) = 1$, if $x_i \in B$ and $h_B(x_i) = -1$, if $x_i \notin B$

$$\begin{aligned} \sum_{i \in P} a_i x_i &= \sum_{j \in N} |a_j| x_j \\ h_B \left(\sum_{i \in P} a_i x_i \right) &= \text{sign} \left(\left\langle w_B, \sum_{i \in P} a_i x_i \right\rangle \right) = \text{sign} \left(\sum_{i \in P} a_i \langle w_B, x_i \rangle \right) = 1 \\ h_B \left(\sum_{j \in N} |a_j| x_j \right) &= \text{sign} \left(\left\langle w_B, \sum_{j \in N} |a_j| x_j \right\rangle \right) = \text{sign} \left(\sum_{j \in N} |a_j| \langle w_B, x_j \rangle \right) = -1 \end{aligned}$$

So, this is a contradiction.

$\text{VCdim}(\mathcal{H}S_0^n)$

Proof:

1st part – show that $\text{VCdim}(\mathcal{H}S_0^n) \geq n$

$A = \{e_1, e_2, \dots, e_n\}$, the orthonormal basis of $\mathbf{R}^n$ is shattered by $\mathcal{H}S_0^n$.

2nd part – show that $\text{VCdim}(\mathcal{H}S_0^n) < n + 1$

Any set $A = \{x_1, x_2, \dots, x_{n+1}\}$ of $n + 1$ points in $\mathbf{R}^n$ cannot be shattered by $\mathcal{H}S_0^n$. Provide an algebraic proof, based on the fact that $\{x_1, x_2, \dots, x_{n+1}\}$ are linearly dependent in $\mathbf{R}^n$.

So, $\text{VCdim}(\mathcal{H}S_0^n) = n$

Similarly, it can be shown that $\text{VCdim}(\mathcal{H}S^n) = n + 1$

The fundamental theorem of statistical learning

The fundamental theorem of statistical learning

Theorem (The Fundamental Theorem of Statistical Learning).

Let $\mathcal{H}$ be a hypothesis class of functions from a domain $\mathcal{X}$ to $\{0,1\}$ and let the loss function be the 0–1 loss. Then, the following statements are equivalent:

1. $\mathcal{H}$ has the uniform convergence property.
2. Any ERM rule is a successful agnostic PAC learner for $\mathcal{H}$.
3. $\mathcal{H}$ is agnostic PAC learnable.
4. $\mathcal{H}$ is PAC learnable.
5. Any ERM rule is a successful PAC learner for $\mathcal{H}$.
6. $\mathcal{H}$ has a finite VC-dimension.

A finite VC- dimension guarantees learnability. Hence, the VC-dimension characterizes PAC learnability.

Proof

1. $\mathcal{H}$ has the uniform convergence property.
 2. Any ERM rule is a successful agnostic PAC learner for $\mathcal{H}$.
 3. $\mathcal{H}$ is agnostic PAC learnable.
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 6. $\mathcal{H}$ has a finite VC-dimension.
- 

Proof:

1 $\rightarrow$ 2 follows from lecture 4: uniform convergence property $\rightarrow$ every sample S is ε -representative $\rightarrow$ ERM is a successful agnostic PAC learner

2 $\rightarrow$ 3, 3 $\rightarrow$ 4 (lecture 5), 2 $\rightarrow$ 5 follow immediately from the definition

4 $\rightarrow$ 6 (lecture 5), 5 $\rightarrow$ 6 – follow from the No-Free Lunch theorem

Need to prove 6 $\rightarrow$ 1 (the hardest part)

Remember – lecture 4: uniform convergence property

Definition (*uniform convergence*)

A hypothesis class $\mathcal{H}$ has the *uniform convergence property* wrt a domain Z , loss function ℓ if:

- there exists a function $m_H^{UC} : (0,1)^2 \rightarrow \mathbb{N}$
- such that for all $(\epsilon, \delta) \in (0,1)^2$
- and for any probability distribution $\mathcal{D}$ over Z

if S is a sample of $m \geq m_H^{UC}(\epsilon, \delta)$ examples drawn i.i.d. according to $\mathcal{D}$, then, with probability of at least $1 - \delta$, S is ϵ -representative.

Definition (ϵ – representative sample)

A sample S is called ϵ – representative wrt domain Z , hypothesis class $\mathcal{H}$, loss function ℓ and distribution $\mathcal{D}$ if:

$$\forall h \in \mathcal{H}, |L_S(h) - L_{\mathcal{D}}(h)| \leq \epsilon.$$

Lemma

Let S be a sample that is $\epsilon/2$ – representative wrt domain Z , hypothesis class $\mathcal{H}$, loss function ℓ and distribution $\mathcal{D}$. Then any output of $\text{ERM}_{\mathcal{H}}(S)$ i.e any $h_S \in \arg\min_h L_S(h)$ satisfies:

$$L_{\mathcal{D}}(A(S)) \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \epsilon$$

Proof for $6 \rightarrow 1$

We want to prove that finite VC-dimension $\rightarrow$ *uniform convergence property*

Two steps:

1. (Sauer's lemma) If $\text{VCdim}(\mathcal{H}) \leq d < \infty$, then even though $\mathcal{H}$ might be infinite, when restricting it to a finite set $C \subseteq \mathcal{X}$, its “effective” size, $|\mathcal{H}_C|$, is only $O(|C|^d)$. That is, the size of $\mathcal{H}_C$ grows polynomially rather than exponentially with $|C|$.
2. we have shown in lecture 4 that finite hypothesis classes enjoy the uniform convergence property. We generalize this result and show that uniform convergence holds whenever the hypothesis class has a “small effective size.” By “small effective size” we mean classes for which $|\mathcal{H}_C|$ grows polynomially with $|C|$.

The Growth function

Definition

Let $\mathcal{H}$ be a hypothesis class. Then the growth function of $\mathcal{H}$, denoted by $\tau_{\mathcal{H}}$, where $\tau_{\mathcal{H}}: \mathbf{N} \rightarrow \mathbf{N}$, is defined as:

$$\tau_{\mathcal{H}}(m) = \max_{C \subseteq X: |C|=m} |H_C|$$

In other words, $\tau_{\mathcal{H}}(m)$ is the maximum number of different functions from a set C of size m to $\{0,1\}$ that can be obtained by restricting $\mathcal{H}$ to C .

Observation: if $\text{VCdim}(\mathcal{H}) = d$ then for any $m \leq d$ we have $\tau_{\mathcal{H}}(m) = 2^m$. In such cases, $\mathcal{H}$ induces all possible functions from C to $\{0,1\}$.

What happens when m becomes larger than the VC-dimension?

Answer given by the Sauer's lemma: the growth function $\tau_{\mathcal{H}}$ increases polynomially rather than exponentially with m .

The Growth function

Definition

Let $\mathcal{H}$ be a hypothesis class. $\tau_H(m)$ is the maximum number of different functions from a set C of size m to $\{0,1\}$ that can be obtained by restricting $\mathcal{H}$ to C .

$$\tau_H(m) = \max_{C \subseteq X: |C|=m} |H_C|$$

Example: consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line $\mathcal{H}_{\text{thresholds}} = \{h_a: \mathbb{R} \rightarrow \{0, 1\}, h_a(x) = \mathbf{1}_{[x < a]}, a \in \mathbb{R}\}$, $|\mathcal{H}_{\text{thresholds}}| = \infty$. We know that $\text{VCdim}(\mathcal{H}) = 1$.

Consider $C = \{c_1, c_2, \dots, c_m\}$ a set of m points, with $c_i < c_j$. What is $\tau_H(m)$?

The Growth function

Definition

Let $\mathcal{H}$ be a hypothesis class. $\tau_H(m)$ is the maximum number of different functions from a set C of size m to $\{0,1\}$ that can be obtained by restricting $\mathcal{H}$ to C .

$$\tau_H(m) = \max_{C \subseteq X: |C|=m} |H_C|$$

Example:

Consider $C = \{c_1, c_2, \dots, c_m\}$ a set of m points, with $c_i < c_j$. What is $\tau_H(m)$?

$\mathcal{H}_C$ can have at most $m+1$ different functions: take $a_1 < c_1 < a_2 < c_2 < \dots < a_m < c_m < a_{m+1}$, then we will have $|\mathcal{H}_C| = \{h_{a_1}, h_{a_2}, \dots, h_{a_{m+1}}\} = m+1$ as

h_{a_1} labels points $c_1, c_2, \dots, c_m$ with labels $(0, 0, 0, \dots, 0, 0)$

h_{a_2} labels points $c_1, c_2, \dots, c_m$ with labels $(1, 0, 0, \dots, 0, 0)$

h_{a_3} label points $c_1, c_2, \dots, c_m$ with labels $(1, 1, 0, \dots, 0, 0)$

.....

h_{a_m} label points $c_1, c_2, \dots, c_m$ with labels $(1, 1, 1, \dots, 1, 0)$

$h_{a_{m+1}}$ label points $c_1, c_2, \dots, c_m$ with labels $(1, 1, 1, \dots, 1, 1)$

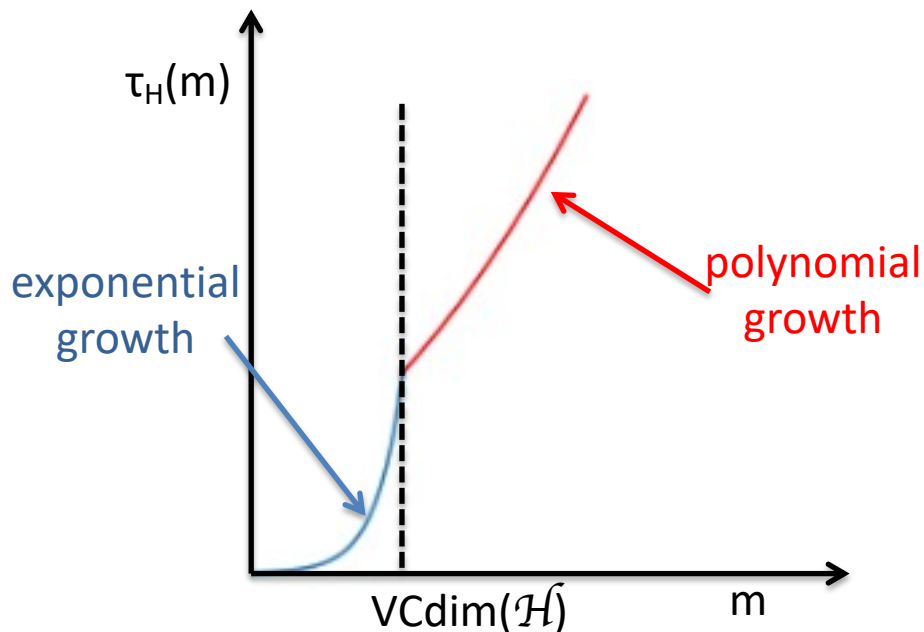
The Sauer's lemma

Lemma (Sauer – Shelah – Perles)

Let $\mathcal{H}$ be a hypothesis class with $\text{VCdim}(\mathcal{H}) \leq d < \infty$. Then, for all m , we have that:

$$\tau_{\mathcal{H}}(m) \leq \sum_{i=0}^d C_m^i$$

In particular, if $m > d + 1$ then $\tau_{\mathcal{H}}(m) \leq (em/d)^d = O(m^d)$



The Sauer's lemma - proof

Lemma (Sauer – Shelah – Perles)

Let $\mathcal{H}$ be a hypothesis class with $\text{VCdim}(\mathcal{H}) \leq d < \infty$. Then, for all m , we have that:

$$\tau_H(m) \leq \sum_{i=0}^d C_m^i$$

In particular, if $m > d + 1$ then $\tau_H(m) \leq (em/d)^d = O(m^d)$

Proof

To prove the lemma it suffices to prove the following stronger claim:

For any $C = \{c_1, c_2, \dots, c_m\}$ we have:

$$|\mathcal{H}_C| \leq |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}|, \text{ for all } \mathcal{H} \text{ a hypothesis class}$$

The reason why this claim is sufficient to prove the lemma is that if $\text{VCdim}(\mathcal{H}) \leq d$ then no set B whose size is larger than d is shattered by $\mathcal{H}$ and therefore:

$$\tau_H(m) = \max_{C \subseteq X: |C|=m} |H_C| \leq \max_{C \subseteq X: |C|=m} |\{B \subseteq C : |B| \leq d\}| \leq \sum_{i=0}^d C_m^i$$

The Sauer's lemma - proof

We will employ induction over the size of C

First step: Fix $\mathcal{H}$ and consider $|C| = 1$.

If $|\mathcal{H}_C| = 1 \leq |\{B \subseteq C: \mathcal{H} \text{ shatters } B\}| = 1$ ($\mathcal{H}$ shatters the empty set).

If $|\mathcal{H}_C| = 2 \leq |\{B \subseteq C: \mathcal{H} \text{ shatters } B\}| = 2$ ($\mathcal{H}$ shatters the empty set and C)

Induction step:

Assume the claim holds for $|C| \leq m$ and prove it for $|C| = m+1$.

Fix $\mathcal{H}$ and consider $C = \{c_1, c_2, \dots, c_m, c_{m+1}\}$ and $C' = \{c_1, c_2, \dots, c_m\}$, so
 $C = C' \cup \{c_{m+1}\}$

Take $Y_0 = \{g: C' \rightarrow \{0, 1\} \mid \text{exists } h \in \mathcal{H} \text{ such}$

that $h(c) = g(c)$ for all $c \in C'$ and $h(c_{m+1}) = 0$

OR $h(c_{m+1}) = 1\}$

So, $Y_0 = \mathcal{H}_C$,

Y_0 →

c_1	c_2	...	c_m	c_{m+1}
1	1	0	1	0
1	1	0	1	1
0	1	1	1	1
1	0	0	1	0
1	0	0	0	1
...	...	...	...	...

The Sauer's lemma - proof

We will employ induction over the size of C

First step: Fix $\mathcal{H}$ and consider $|C| = 1$.

If $|\mathcal{H}_C| = 1 \leq |\{B \subseteq C: \mathcal{H} \text{ shatters } B\}| = 1$ ($\mathcal{H}$ shatters the empty set).

If $|\mathcal{H}_C| = 2 \leq |\{B \subseteq C: \mathcal{H} \text{ shatters } B\}| = 2$ ($\mathcal{H}$ shatters the empty set and C)

Induction step:

Assume the claim holds for $|C| \leq m$ and prove it for $|C| = m+1$.

Fix $\mathcal{H}$ and consider $C = \{c_1, c_2, \dots, c_m, c_{m+1}\}$ and $C' = \{c_1, c_2, \dots, c_m\}$.

Take $Y_0 = \{g: C' \rightarrow \{0, 1\} \mid \text{exists } h \in \mathcal{H} \text{ such that } h(c) = g(c) \text{ for all } c \in C' \text{ and } h(c_{m+1}) = 0 \text{ OR } h(c_{m+1}) = 1\} = \mathcal{H}_C$,

If there exists two different function h_1 and h_2 in $\mathcal{H}$ that agree with g on C' then they will disagree on c_{m+1} : $h_1(c_{m+1}) \neq h_2(c_{m+1})$. They are two different functions in $\mathcal{H}$ but they will be counted only once in Y_0 .

The Sauer's lemma - proof

Take $Y_0 = \{g: C' \rightarrow \{0, 1\} \mid \text{exists } h \in \mathcal{H} \text{ such that } h(c) = g(c) \text{ for all } c \in C' \text{ and } h(c_{m+1}) = 0 \text{ OR } h(c_{m+1}) = 1\} = \mathcal{H}_C$,

Take $Y_1 = \{g: C' \rightarrow \{0, 1\} \mid \text{exists } h_1, h_2 \in \mathcal{H} \text{ such that } h_1(c) = g(c) \text{ for all } c \in C' \text{ and } h_1(c_{m+1}) = 0 \text{ AND } h_2(c) = g(c) \text{ for all } c \in C' \text{ and } h_2(c_{m+1}) = 1\}$

	c_1	c_2	...	c_m	c_{m+1}	
	1	1	0	1	0	h_1
Y_1	1	1	0	1	1	h_2
	0	1	1	1	1	
	1	0	0	1	0	
Y_0	1	0	0	0	1	
	...	...	...	...	...	

The Sauer's lemma - proof

Take $Y_0 = \{g: C' \rightarrow \{0, 1\} \mid \text{exists } h \in \mathcal{H} \text{ such that } h(c) = g(c) \text{ for all } c \in C' \text{ and } h(c_{m+1}) = 0 \text{ OR } h(c_{m+1}) = 1\} = \mathcal{H}_C$,

Take $Y_1 = \{g: C' \rightarrow \{0, 1\} \mid \text{exists } h_1, h_2 \in \mathcal{H} \text{ such that } h_1(c) = g(c) \text{ for all } c \in C' \text{ and } h_1(c_{m+1}) = 0 \text{ AND } h_2(c) = g(c) \text{ for all } c \in C' \text{ and } h_2(c_{m+1}) = 1\}$

We have that $Y_1 \subseteq Y_0$

Y_1 contains only those restriction h_C that come from two different functions h_1 and h_2 from $\mathcal{H}$

Y_0 might contain restrictions h_C that come from a single h from H .

For simplicity let's assume that $C = \mathcal{X}$, $\mathcal{X}$ is the domain of $\mathcal{H}$.

We have that $|H| = |Y_0| + |Y_1|$

The Sauer's lemma - proof

Take $Y_0 = \{g: C' \rightarrow \{0, 1\} \mid \text{exists } h \in \mathcal{H} \text{ such that } h(c) = g(c) \text{ for all } c \in C' \text{ and } h(c_{m+1}) = 0 \text{ OR } h(c_{m+1}) = 1\} = \mathcal{H}_C$,

Take $Y_1 = \{g: C' \rightarrow \{0, 1\} \mid \text{exists } h_1, h_2 \in \mathcal{H} \text{ such that } h_1(c) = g(c) \text{ for all } c \in C' \text{ and } h_1(c_{m+1}) = 0 \text{ AND } h_2(c) = g(c) \text{ for all } c \in C' \text{ and } h_2(c_{m+1}) = 1\}$

	c_1	c_2	...	c_m	c_{m+1}
$Y_1 \rightarrow$	1	1	0	1	0
	1	1	0	1	1
$Y_0 \rightarrow$	0	1	1	1	1
	1	0	0	1	0
	1	0	0	0	1
	...	...	...	...	...

h_1
 h_2

The Sauer's lemma - proof

Now, we will apply our induction hypothesis on Y_0

$$|Y_0| = |\mathcal{H}_{C'}| \leq |\{B \subseteq C' : \mathcal{H} \text{ shatters } B\}| = |\{B \subseteq C : \mathcal{H} \text{ shatters } B \text{ and } c_{m+1} \notin B\}|$$

Take $\mathcal{H}' = \{h_1 \in \mathcal{H} \text{ such that there exists } h_2 \in \mathcal{H} \text{ s. t. for all } c \in C' \text{ we have } h_1(c) = h_2(c) \text{ but } h_1(c_{m+1}) \neq h_2(c_{m+1})\}$

Then $Y_1 = \mathcal{H}'_{C'} = \text{set of function on } C' \text{ with two extensions on } c_{m+1}$

Use the induction hypothesis here, on Y_1 :

$$|Y_1| = |\mathcal{H}'_{C'}| \leq |\{B \subseteq C' : \mathcal{H}' \text{ shatters } B\}| = |\{B \subseteq C : \mathcal{H} \text{ shatters } B \text{ and } c_{m+1} \in B\}|$$

So, we have that $|\mathcal{H}| = |\mathcal{H}_C| \leq |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}|$

$\tau_{\mathcal{H}}$ grows polynomially

Corollary

Let H be a hypothesis class with $\text{VCdim}(H) = d$. Then for all $m \geq d$:

$$\tau_H(m) \leq \left(\frac{em}{d}\right)^d = O(m^d)$$

Proof:

From the Sauer lemma we have:

$$\tau_H(m) \leq \sum_{i=0}^d C_m^i \leq \sum_{i=0}^d \left(C_m^i \times \left(\frac{m}{d}\right)^{d-i} \right) \leq \sum_{i=0}^m \left(C_m^i \times \left(\frac{m}{d}\right)^{d-i} \right) = \left(\frac{m}{d}\right)^d \sum_{i=0}^m \left(C_m^i \times \left(\frac{d}{m}\right)^i \right)$$

$\uparrow$ $m \geq d$
 $\uparrow$ $m \geq d$

$$\tau_H(m) \leq \left(\frac{m}{d}\right)^d \sum_{i=0}^m \left(C_m^i \times \left(\frac{d}{m}\right)^i \right) = \left(\frac{m}{d}\right)^d \left(1 + \frac{d}{m} \right)^m \leq \left(\frac{m}{d}\right)^d \left(e^{\frac{d}{m}} \right)^m = \left(\frac{em}{d}\right)^d$$

$\uparrow$ Newton's binomial formula
 $\uparrow$ $1-x \leq e^{-x}$

Proof for $6 \rightarrow 1$

We want to prove that finite VC-dimension $\rightarrow$ *uniform convergence property*

Two steps:

1. (Sauer's lemma) If $\text{VCdim}(\mathcal{H}) = d < \infty$, then even though $\mathcal{H}$ might be infinite, when restricting it to a finite set $C \subseteq \mathcal{X}$, its “effective” size, $|\mathcal{H}_C|$, is only $O(|C|^d)$. That is, the size of $\mathcal{H}_C$ grows polynomially rather than exponentially with $|C|$.
2. we have shown in lecture 4 that finite hypothesis classes enjoy the uniform convergence property. We generalize this result and show that uniform convergence holds whenever the hypothesis class has a “small effective size.” By “small effective size” we mean classes for which $|\mathcal{H}_C|$ grows polynomially with $|C|$.

Uniform converge holds for $\mathcal{H}$ with small effective size

Theorem

Let $\mathcal{H}$ be a class and let $\tau_{\mathcal{H}}$ be its growth function. Then, for every $\mathcal{D}$ and every $\delta \in (0,1)$, with probability of at least $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$ we have:

$$|L_D(h) - L_S(h)| \leq \frac{4 + \sqrt{\log(\tau_H(2m))}}{\delta \sqrt{2m}}$$

Proof:

- in the book, is beyond the scope of this lecture

Proof for $6 \rightarrow 1$

We want to prove that finite VC-dimension $\rightarrow$ *uniform convergence property*.

Combine the last result with Sauer lemma: $\tau_{\mathcal{H}}(m) \leq (em/d)^d = O(m^d)$ to obtain:
for every $\mathcal{D}$ and every $\delta \in (0,1)$, with probability of at least $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$ we have:

$$|L_D(h) - L_S(h)| \leq \frac{4 + \sqrt{\log(\tau_H(2m))}}{\delta \sqrt{2m}} \leq \frac{4 + \sqrt{d \log(2em/d)}}{\delta \sqrt{2m}} \leq \frac{2\sqrt{d \log(2em/d)}}{\delta \sqrt{2m}}$$

$\uparrow$ Sauer lemma $\tau_H(2m) \leq (2em/d)^d$
 $\uparrow$ consider m such that $4^2 \leq d \log(2em/d)$

$$|L_D(h) - L_S(h)| \leq \frac{1}{\delta} \frac{\sqrt{2d \log(2em/d)}}{\sqrt{m}} < \varepsilon$$

This leads (see the calculation in the book) to:

$$m \geq 4 \frac{2d}{(\delta \varepsilon^2)} \log\left(\frac{2d}{\delta \varepsilon^2}\right) + \frac{4d \log(2\varepsilon/d)}{(\delta \varepsilon^2)}$$

Proof for $6 \rightarrow 1$

We want to prove that finite VC-dimension $\rightarrow$ *uniform convergence property*.

for every $\mathcal{D}$ and every $\delta \in (0,1)$, with probability of at least $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$ we have that if:

$$m \geq 4 \frac{2d}{(\delta\epsilon^2)} \log\left(\frac{2d}{\delta\epsilon^2}\right) + \frac{4d \log(2\epsilon / d)}{(\delta\epsilon^2)}$$

then the sample S is ϵ -representative

$$|L_D(h) - L_S(h)| \leq \frac{1}{\delta} \frac{\sqrt{2d \log(2em / d)}}{\sqrt{m}} < \epsilon$$

So, we have that: $m_H^{UC}(\epsilon, \delta) \leq 4 \frac{2d}{(\delta\epsilon^2)} \log\left(\frac{2d}{\delta\epsilon^2}\right) + \frac{4d \log(2\epsilon / d)}{(\delta\epsilon^2)}$

The derived bound is not the tightest possible, there exist another bound much tighter (see next).

The fundamental theorem of statistical learning – quantitative version

Theorem

Let $\mathcal{H}$ be a hypothesis class of functions from a domain $\mathcal{X}$ to $\{0,1\}$ and let the loss function be the 0–1 loss. Assume that $\text{VCdim}(\mathcal{H}) = d < \infty$. Then, there are absolute constants C_1, C_2 such that:

1. $\mathcal{H}$ has the uniform convergence property with sample complexity:

$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}^{UC}(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2}$$

2. $\mathcal{H}$ is agnostic PAC learnable with sample complexity:

$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2}$$

3. $\mathcal{H}$ is PAC learnable with sample complexity:

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$$

The VC dimension determines (along with ϵ, δ) the samples complexities of learning a class. It gives us a lower and an upper bound.

Intuition for deriving the lower bounds

The PAC case (realizable case)

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$$

Pick a set $A = \{x_1, x_2, \dots, x_d\}$ of size d ($=VCdim(\mathcal{H})$) that is shattered by $\mathcal{H}$. Choose the following (adversarial) probability distribution $\mathcal{D}$ over $\mathcal{X}$:

$\mathcal{D}(x_1) = 1-4\epsilon$, $\mathcal{D}(x_i) = 4\epsilon/(d-1)$, $i = 2, 3, \dots, d$, $\mathcal{D}(x) = 0$, for all x in $\mathcal{X} \setminus A$

By the No Free Lunch theorem as long as a sample S hits $B = \{x_2, \dots, x_d\}$ at most $(d-1)/2$ times, the probability of making an error over B is $\geq 1/4$. This happens because we see less than half of the domain B points. So, our expected error with respect to $\mathcal{D}$ is $4\epsilon/4 = \epsilon$.

If the sample S has size m , then roughly $4m\epsilon$ points will hit $B = \{x_2, \dots, x_d\}$. So, to make less than ϵ errors we need to have $4m\epsilon > (d-1)/2$, $m > (d-1)/8\epsilon$

Assignment 1

Assignment 1 – good to know

- 3 problems = 2.7 points + 0.3 points (ex-officio)
- deadline: in about 4 weeks time, Tuesday 5th of May 2026, 23:59
 - late submission policy: maximum 3 days allowed, -10% (= 0.3 points) for each day
 - upload a pdf written in a scientific editor (Word, Latex, LyX) containing your solution here: <https://tinyurl.com/AML-2026-ASSIGNMENT1>
 - *is mandatory that you write your solution with a scientific editor, otherwise your solution would not be taken into account*
 - you can insert drawings for your proofs
 - after uploading your solution you should get a confirmation email; if not send me an email.
- for every problem write clear explanations, proofs to justify your answer (if you write just some indications you will not get too many points)
- do not share/copy the solution with/from your colleagues: you + your colleague/s will get 0 points

Assignment 1

Deadline: Tuesday, 5th of May 2026

Upload your solutions at: <https://tinyurl.com/AML-2026-ASSIGNMENT1>

1. **(0.3 points)** What is the maximum value of the natural even number n , $n = 2m$, such that there exists a hypothesis class $\mathcal{H}$ with n elements that shatters a set C of $m = \frac{n}{2}$ points? Give an example of such an $\mathcal{H}$ and C . Justify your answer.

2. **(1.2 points)** Consider $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{H}_3$, where:

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[x \geq a]}(x) = \mathbf{1}_{[a, +\infty)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[x < a]}(x) = \mathbf{1}_{(-\infty, a)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_{a,b} : \mathbb{R} \rightarrow \{0, 1\} \mid h_{a,b}(x) = \mathbf{1}_{[a \leq x \leq b]}(x) = \mathbf{1}_{[a,b]}(x), a, b \in \mathbb{R}\}.$$

Consider the realizability assumption.

a) Compute $\text{VCdim}(\mathcal{H})$.

b) Give an algorithm A and determine an upper bound on the sample complexity $m_{\mathcal{H}}(\epsilon, \delta)$ such that the definition of PAC-learnability is satisfied.

3. **(1.2 points)** Usually, there is a correlation between the number of parameters of a hypothesis class $\mathcal{H}$ and its VC-dimension. Consider the following hypothesis classes:

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[a, a+1]}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_b : \mathbb{R} \rightarrow \{0, 1\} \mid h_b(x) = \mathbf{1}_{[b, b+1] \cup [b+2, b+4]}(x), b \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_c : \mathbb{R} \rightarrow \{0, 1\} \mid h_c(x) = \mathbf{1}_{[c, c+1] \cup [c+2, c+4] \cup [c+6, c+9]}(x), c \in \mathbb{R}\}.$$

Consider the realizability assumption. Compute $\text{VCdim}(\mathcal{H}_1)$, $\text{VCdim}(\mathcal{H}_2)$, $\text{VCdim}(\mathcal{H}_3)$.

Ex-officio: 0.3 points